

Spiking neural networks for EEG signal analysis: From theory to practice

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ABSTRACT

The intricate and efficient information processing of the human brain, driven by spiking neural interactions, has led to the development of spiking neural networks (SNNs) as a cutting-edge neural network paradigm. Unlike traditional artificial neural networks (ANNs) that use continuous values, SNNs emulate the brain's spiking mechanisms, offering enhanced temporal information processing and computational efficiency. This review addresses the critical gap between theoretical advancements and practical applications of SNNs in EEG signal analysis. We provide a comprehensive examination of recent SNN methodologies and their application to EEG signals, highlighting their potential benefits over conventional deep learning approaches. The review encompasses foundational knowledge of SNNs, detailed implementation strategies for EEG analysis, and challenges inherent to SNN-based methods. Practical guidance is provided through step-by-step instructions and accessible code available on GitHub, aimed at facilitating researchers' adoption of these techniques. Additionally, we explore emerging trends and future research directions, emphasizing the potential of SNNs to advance brain-computer interfaces and neurofeedback systems. This paper serves as a valuable resource for bridging the gap between theoretical developments in SNNs and their practical implementation in EEG signal analysis.

1. Introduction

The human brain, comprising approximately 90 billion neurons interconnected at microstructural levels (Azevedo et al., 2009), operates as a sophisticated and efficient system for processing and transmitting information in real time. Communication between neurons occurs through electrical impulses known as spikes. Understanding how the brain acquires, transmits, and efficiently processes information has been a fundamental pursuit in neuroscience research.

Neural networks, or artificial neural networks (ANNs), were inspired by biological neural networks (Hinton et al., 2012; Hinton & Salakhutdinov, 2006). A neural network model consists of interconnected neurons, mimicking the brain's structure. The development of neural networks can be divided into three key phases. The first phase began in 1958 with the introduction of the "Perceptron" (Rosenblatt, 1958), a

model designed to simulate human perception but limited to linearly separable problems (Schmidhuber, 2015). The second phase emerged in 1989 when (Hornik et al., 1989) demonstrated that multi-layer NNs could approximate complex functions. Since 2006, deep neural networks (DNNs) have driven the development of artificial functions. Despite many successes, ANNs remain computationally expensive and lack full biological plausibility, differing significantly from human learning mechanisms (LeCun et al., 2015).

Spiking neural networks (SNNs) (Maass, 1997), inspired by the brain and its underlying mechanisms, have become an appealing alternative to ANNs. Their complex spatio-temporal dynamics, diverse coding strategies, and event-driven advantages position them as promising next-generation neural network paradigms (Taherkhani et al., 2020). Unlike ANNs, SNNs process information through binary impulses, closely mimicking the brain's information processing. This unique characteris-

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tic makes SNNs highly suitable for ultra-low-power, event-driven neuromorphic hardware (Zhang et al., 2021b).

As SNNs draw inspiration from both neuroscience and advances in ANNs, their unique capabilities position them as powerful tools for brain signal analysis (Rathi et al., 2023). This calls for a more detailed exploration of the current state of SNN research, especially regarding its application to processing brain signals. There are various methods for measuring brain activity (Tang et al., 2023), including electroencephalography (EEG), electrocorticography (ECoG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), etc. Among these, EEG stands out due to its high temporal resolution, non-invasiveness, and cost-effectiveness, making it a preferred choice in both clinical and research settings (He et al., 2023).

Traditional EEG signal processing relies heavily on machine learning algorithms to handle complex, non-linear, and non-stationary data (Duong et al., 2022; Hosseini et al., 2021). Although these methods have shown effectiveness in various applications, they often demand substantial computational resources and do not fully exploit the temporal dynamics inherent in EEG signals (Chen & Ran, 2019; Prochnow et al., 2024). Moreover, conventional machine learning approaches process continuous values, making them inherently mismatched for the discrete, event-driven nature of neuronal activity (Luo, 2021). SNNs, by contrast, are naturally suited for this task, as their event-driven architecture directly mimics the brain's spiking dynamics.

SNNs encode information as discrete spikes to represent temporal patterns with high precision (Petro et al., 2020). Such spiking mechanism conforms well to the inherently temporal and event-driven nature of EEG signals (Virgilio G. et al., 2020). A key advantage of SNNs is computational efficiency, critical for real-time EEG applications like brain-computer interfaces (BCIs) and neurofeedback systems. With sparse coding and event-driven computation, SNNs involve much fewer operations than the dense matrix multiplications in most deep learning methods, which means lower power consumption and processing latency required by portable BCIs (Yan et al., 2022). Additionally, since EEG signals reflect neuronal activity, SNNs provide a biologically plausible framework for interpreting neural processes (Cai et al., 2023a, 2024; Carlos D. Virgilio et al., 2020), advancing insights into cognitive functions and neurological disorders (Cherian & Kanaga E, 2024; Jiao et al., 2024).

To fully benefit from the desired properties of SNNs for EEG signal analysis, it is necessary to understand the underlying working mechanisms and the state-of-the-art. Recently, a literature review (Xiaoxue et al., 2023) covers how SNNs can bridge computational neuroscience with medical data, including EEG, electrocardiography (ECG), electromyography (EMG), and MRI, for early disease detection. It provides insights into advancements in signal classification and disease diagnosis, comparing SNNs with traditional neural networks. Additionally, a mini-review (Zhan et al., 2021) summarizes the use of SNNs in four EEG signal processing tasks, particularly through NeuCube-based methods (Kasabov, 2014). A more recent review has explored the potential of SNNs for epileptic seizure detection based on EEG signals (Cherian & Kanaga E, 2024). While these reviews provide valuable insights, they often focus on specific techniques but lack the necessary breadth and depth for practical implementation.

There exists a gap between theoretical advancements and practical implementation, underscoring the need for a comprehensive review. This survey article is motivated to bridge the gap between theory and practice and provide a coverage of the state-of-the-art. The contributions of this survey can be summarized as follows:

- We provide an in-depth review of the current state of SNN-based EEG signal analysis, focusing on recent advancements and the evolution of SNN methodologies within this domain.
- We provide detailed, step-by-step instructions for implementing SNNs in EEG analysis tasks. Additionally, we have made all code available on GitHub to empower researchers to apply these SNN techniques in their EEG-related projects easily.

- We explore the challenges inherent in SNN-based EEG analysis approaches and discuss emerging trends and research directions, thus paving the way for further advancements and investigations in this field.

The rest of this survey is organized as follows. Section 2 covers the fundamentals of SNNs. In Section 3, we provide the basics of EEG signals and analysis to set the stage for subsequent discussions. Section 4 introduces different spiking neural architectures and learning algorithms in the context of EEG signal processing. In Section 5, we study the practical steps through a project workshop. Section 6 discusses emerging trends and future research directions, after which Section 7 concludes this survey.

2. Fundamentals of spiking neural networks

Let's start by providing an overview of the core concepts of SNNs, including neural encoding methods and operational mechanisms of networks, that are relevant to EEG signal analysis. We also highlight how SNNs differ from traditional neural networks through their unique temporal dynamics and event-driven computation. The unique properties of SNNs make them a favorable alternative for applications in EEG signal analysis.

2.1. Neural encoding

Due to the discrete nature of spiking neural processing, it presents many challenges as to how to adequately present analog signals, e.g., EEG signals, to SNNs. In the early studies, analog signals are typically preprocessed to become compatible with SNNs (Gupta et al., 2013; Yilmaz et al., 2020). Such preprocessing may introduce noise and distortion, which affect the quality of the input signals (Wu et al., 2018a). The continuous nature of analog signals is inherently incompatible with binary spiking processes, leading to inefficiencies during both model training and inference (Yang et al., 2022). Such incompatibility can result in significant computational demands and processing delays, particularly in applications requiring real-time or low-latency responses (Wang et al., 2024b). In sum, it is critical to effectively encode analog signals into spatiotemporal spike trains so as to fully take advantage of SNNs (Pan et al., 2021a).

The most commonly used spatiotemporal encoding schemes can be generally categorized into two types: rate coding (Al-Hamid & Kim, 2020) and temporal coding (Zhou et al., 2021). As illustrated in Fig. 1, rate coding abstracts information by measuring the frequency of spikes within a given time window, focusing on the overall spike rate rather than the specific timing of each spike (Kim et al., 2022; Wang et al., 2023b). In contrast, temporal coding takes into account both the number of spikes and their precise timing, thus encoding information based on the exact moments of spike emissions (Petro et al., 2020; Yu et al., 2022).

2.1.1. Rate encoding

Rate encoding (Gerstner & Kistler, 2002), or rate coding, is a coding mechanism that follows the principle with which the frequency of action potentials in spiking neurons relays information. The informational content is primarily gauged by the spike rate per time unit, with elevated frequencies indicating intensified stimuli. This concept is mathematically formulated as follows,

$$F = \mathcal{F}\left(\frac{N}{T}\right), \quad (1)$$

where F represents the frequency of spiking, N is the number of spikes observed during a time window T , and $\mathcal{F}$ denotes a mapping function, typically following Poisson or Gaussian distributions. Specifically, the Poisson model (Adrian, 1926) treats neural firing as a Poisson process,

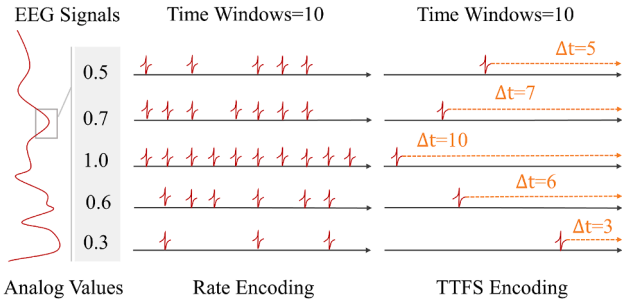


Fig. 1. The rate encoding method represents the analog values of signals through the firing frequency of spike trains within a time window, where the distribution is random and uncertain. Conversely, the Time-to-First-Spike (TTFS) method, a prevalent form of temporal encoding, represents the analog values based on the timing of the first spike, with larger values prompting earlier spikes.

where spikes occur independently at a constant rate over short intervals. The Poisson encoding formula is given by:

$$P(N = n) = \frac{\lambda^n e^{-\lambda}}{n!}, \quad (2)$$

Here, $P(N = n)$ represents the probability of observing n spikes within a given interval, λ is the spike count, e is the base of the natural logarithm, and $n!$ is the factorial of n .

Although rate encoding simplifies data representation through basic functional mappings, it is not the most computationally efficient method. This strategy relies on monitoring pulse frequencies within a time window, where higher frequencies lead to increased energy consumption (Kim et al., 2022; Wang et al., 2023b). Moreover, the process of encoding and decoding information requires spiking statistics over a time window, which can constrain the system's ability to process rapidly changing information effectively (Guo et al., 2021).

2.1.2. Temporal encoding

Unlike rate coding, temporal encoding, or temporal coding, makes use of the precise timing of spikes, in addition to their frequency (Golisch & Meister, 2008). The timing relative to a universal reference carries information, enabling neural networks to use temporal differences between spikes for encoding (Guo et al., 2021). Temporal encoding provides a robust framework for representing time-sensitive data, and modeling temporal relationships and dynamics. It is particularly advantageous when processing complex temporal patterns (Zhou et al., 2021).

Time-to-first-spike (TTFS) is one temporal coding technique, that emphasizes the timing of the initial spike following a stimulus (Johansson & Birnieks, 2004). In TTFS encoding, the earlier a neuron fires after a stimulus, the more significant the information it conveys. TTFS is particularly effective for encoding stimuli with precise temporal resolution, emphasizing the importance of early neuronal firing in representing stimulus-related information (Li et al., 2024). Phase coding is another temporal coding technique, that uses the timing differences between spikes and a periodic reference oscillation to encode information (Gray et al., 1989). In this method, neurons fire according to the phase of the oscillatory reference signal. The phase relationship of each spike relative to this oscillation encodes important information, allowing for a representation based on phase differences (Pan et al., 2021b). There are also threshold-based temporal encoding methods such as stochastic step-wise noisy threshold model (ST) (Kasabov et al., 2016), and Ternary Asynchronous Encoding (TAE) (Wang et al., 2024a) convert rapidly fluctuating signals into discrete spike trains without altering the sampling rate. These approaches effectively preserve the temporal features of signals.

2.2. Spiking neural architecture

While neural encoding interfaces the input signals, the neuronal models form the backbone of the networks. An understanding of the

models will provide insight into how SNNs emulate biological neural activity and manage information processing.

2.2.1. Neuron models

Neurons in biological systems are the fundamental structural and functional units of the brain's nervous system. A biological neuron can be considered to have three parts: the cell body, dendrites, and axons. The cell body contains the nucleus, which is responsible for maintaining normal cellular functions; dendrites receive signals; and axons transmit signals, connecting to other neurons through synapses.

Spiking neurons in SNNs and artificial neurons in ANNs both are motivated by our understanding of biological neurons. Unlike the simple weighted operations of artificial neurons, spiking neurons process information through changes in membrane potential, synaptic transmission, and spike generation (Tavanaei et al., 2019). When the input signal exceeds a threshold, a spike is produced, and the membrane potential is reset. This mode of operation gives SNNs a distinct advantage in handling temporal sequence data and simulating the dynamic characteristics of the brain and its biological plausibility. The event-driven property naturally leads to operational efficiency and reduces power consumption (Ghosh-Dastidar & Adeli, 2009b).

The Leaky Integrate-and-Fire (LIF) model (Dayan & Abbott, 2001) is a classic and widely adopted spiking neuron model. It uses a simple differential equation to describe changes in the membrane potential of neurons, simulating how neurons accumulate input signals and generate spikes while considering the natural leakage of the membrane potential, making its behavior closer to that of biological neurons. In the LIF model, the membrane potential gradually accumulates input current. When the neuron receives synaptic input currents, the membrane potential rises. Simultaneously, the membrane potential decays over time through an exponential decay function. When the membrane potential reaches a preset threshold, the neuron generates a spike (action potential) and then resets to a lower value, typically the resting potential. Its mathematical description is:

$$\tau_m \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + RI(t), \quad (3)$$

where $V(t)$ is the membrane potential, changing over time; τ_m is the membrane time constant, representing the rate of decay of the membrane potential; V_{rest} is the resting potential; R is the resistance to the input current; $I(t)$ is the input current varying over time. When $V(t)$ reaches the threshold, the neuron fires and resets the membrane potential to V_{rest} .

The LIF model emulates the dynamic behavior of biological neurons in a straightforward and effective manner. Other common spiking neuron models include the Izhikevich model (Izhikevich, 2003), the Hodgkin-Huxley model (Yamazaki et al., 2022), and the Spike Response Model (SRM) (Jolivet et al., 2003). The Izhikevich model can simulate a variety of complex neuron firing patterns, combining biological realism and computational efficiency. The Hodgkin-Huxley model accurately simulates neuron behavior through the dynamics of currents and ion channels, though with a higher computational complexity. The SRM model calculates the membrane potential based on the timing of input spikes and the spike response function, offering a good balance of complexity and flexibility.

2.2.2. Network architectures

An SNN consists of a number of spiking neurons that are arranged in an appropriate manner, and interact with each other during training and inference. We provide an overview of various network designs that represent the advances in the field, as shown in Fig. 2. Generally, these architectures can be categorized into five types: feedforward, residual, attention-based models, Transformer-based, and large language models.

Early spiking neural networks predominantly draw inspiration from the traditional perceptron, a feedforward neural network architecture. Bi and Poo (1998) established the foundation for spike-timing-dependent plasticity (STDP), a critical mechanism underlying learning

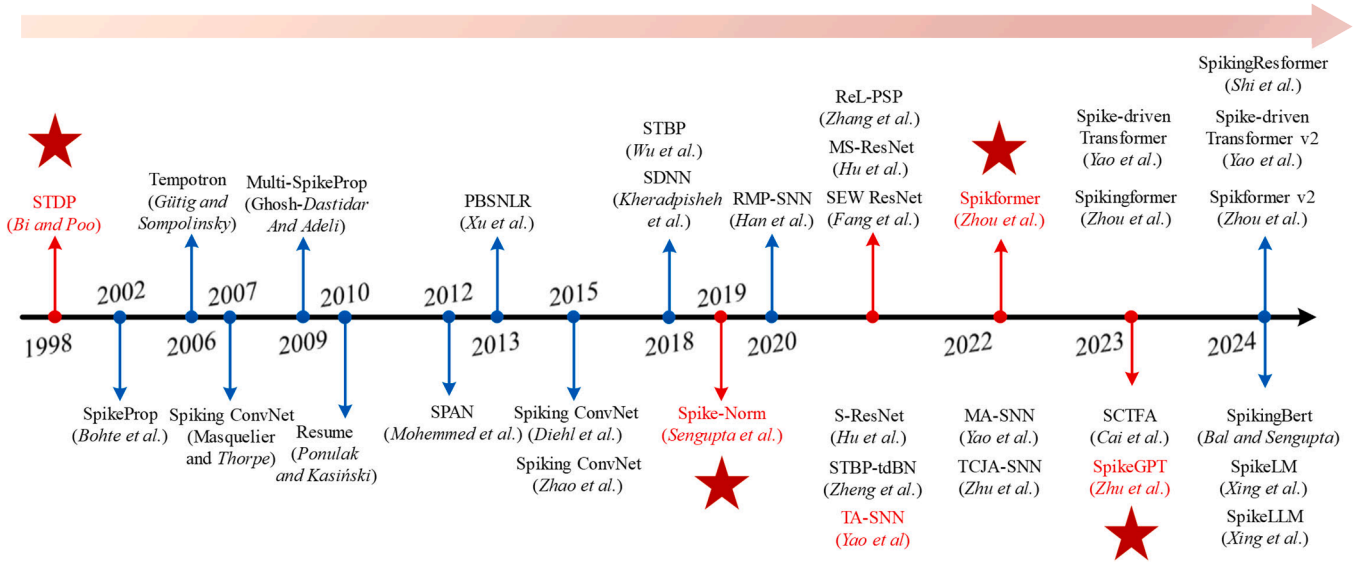


Fig. 2. The evolution of spiking neural network architecture in five stages: feedforward, residual, attention-based, Transformer-based, and large language models. A red star marks the model seen as a significant landmark.

and memory, inspiring extensive research into its role in neural computation, circuit dynamics, and neurological disorders (Bohte et al., 2002; Ghosh-Dastidar & Adeli, 2009a; Gütig & Sompolinsky, 2006; Masquelier & Thorpe, 2007; Mohammed et al., 2012; Ponulak & Kasiński, 2010). These early SNNs typically consisted of multiple fully connected or convolutional layers arranged in a sequential manner, with information flowing directly from the input layer through the hidden layers to the output layer, without any feedback loops or recurrent connections (Diehl et al., 2015; Han et al., 2020; Kheradpisheh et al., 2018; LeCun et al., 2015; Wu et al., 2018b; Xu et al., 2013; Zhang et al., 2021b; Zhao et al., 2015). For instance, SpikeProp (Bohte et al., 2002) and Multi-SpikeProp (Ghosh-Dastidar & Adeli, 2009a) employed a standard three-layer architecture comprising input, hidden, and output layers. There were also studies of spiking convolutional layers that were used for feature extraction (Diehl et al., 2015; Kheradpisheh et al., 2018; Masquelier & Thorpe, 2007; Zhang et al., 2021b; Zhao et al., 2015), demonstrating their effectiveness across multiple tasks.

To overcome the vanishing gradient problem during training in deep neural network architecture, residual network structures were developed (He et al., 2016). The extension of this idea to SNN Sengupta et al. (2019) marked a significant milestone for deep architecture (Fang et al., 2021; Hu et al., 2024, 2021b; Zheng et al., 2021). For example, Zheng et al. (2021) employed threshold-dependent batch normalization for residual connections, successfully training spiking versions of ResNet-34 and ResNet-50. However, Fang et al. (2021) found that traditional spiking residual networks are not suitable for all neuron models to achieve identity mapping. To address this, they introduced the Spike Element-Wise (SEW) method, successfully training a convolutional SNN with over 100 layers.

Due to the dynamic characteristics of spiking neurons, it is believed that they have the ability to handle long sequence tasks. Some researchers (Chen et al., 2024b; Fang et al., 2021; Shen et al., 2024) have attempted to align the timesteps of SNNs with those in sequential tasks to further reduce energy consumption. Yin et al. (2021) introduce adaptive factors into vanilla LIF neurons, enabling SNNs to achieve performance competitive with ANNs. Moreover, Zhang et al. (2024) analyze the gradient explosion and vanishing issues in LIF neurons during sequential learning and improved SNN performance using a dual-compartment neuron design. Zheng et al. (2024) demonstrate through temporal XOR experiments, the significance of decay factor design in sequential tasks and propose the DH-LIF neuron to effectively

balance computational complexity and energy efficiency. Collectively, these studies provide compelling evidence for the feasibility and effectiveness of SNNs in sequence learning tasks.

Recent advances in attention mechanisms, known for their advantages in sequence learning and parallel computing (Devlin et al., 2018) motivated a similar implementation in SNNs. In particular, Yao et al. (2021) and Zhou et al. (2022) introduced temporal and spatial attention mechanisms into SNNs, respectively. The idea of attention was further extended to multi-dimensional attention mechanisms, which yielded promising results on neuromorphic datasets (Cai et al., 2023b; Yao et al., 2023; Zhu et al., 2024).

Building on the success of spiking attention mechanisms, transformer architecture was also studied in SNNs intensively (Shi et al., 2024; Yao et al., 2024a,b; Zhou et al., 2024, 2022). In particular, Zhou et al. (2022) introduced Spiking Transformer (Spikformer), which leverages both self-attention capability and biological properties in the networks. The transformer-based spiking networks have achieved high accuracy on complex tasks, such as those presented by ImageNet, marking a significant milestone in the evolution of SNNs.

The advent of large language models (LLMs) also inspired several studies on spiking counterparts, which showed the potential of large-scale spiking networks in tackling complex tasks such as natural language understanding and generation (Bal & Sengupta, 2024; Xing et al., 2024a,b; Zhu et al., 2023).

2.3. Spiking neural networks for EEG processing

There have been more studies on traditional ANNs than SNNs. Furthermore, we often use ANNs as a reference system to benchmark SNNs. Our understanding of ANNs is useful for the design of SNNs.

We summarize key aspects of the two types of networks in Table 1, that are relevant for EEG analysis. It is considered that SNNs are more suitable for EEG analysis due to their biologically plausible properties, for instance, event-driven computation, i.e. neurons operate independently, a neuron fires only when it is ready to transmit information; and discrete processing, i.e. EEG signals are the collective signals of thousands of nerve cells, each cell firing short, or discrete, pulses of electricity along its output axon. If we consider a spiking neuron as a brain cell, an SNN resembles a human brain.

Neuron models: The neuron model in an ANN is relatively straightforward, typically formulated as the weighted summation of inputs

Table 1
Comparative analysis of artificial neural networks vs. spiking neural networks in several aspects.

	ANNs	SNNs
Neuron model	Weighted sum followed by an activation function (e.g., ReLU, Sigmoid, Tanh)	Mimics biological neurons with dynamic properties (e.g., membrane potential, spikes emission, refractory periods)
Neural activation	Continuous values	Discrete spikes
Information flow	Primarily synchronous updates layer by layer in spatial domain	Event-driven asynchronous updates in temporal and spatial domains
Biological plausibility	Loosely inspired biologically, emphasizing the mathematical formulation of learning	Closely emulating the behavior of biological neurons, capturing essential aspects like spike generation and transmission to better reflect biological processes
Power consumption	High power consumption due to dense, continuous computation	Low power consumption through sparse, event-driven processing
Hardware compatibility	Widely supported by existing digital hardware, GPUs, and TPUs	Better suited for neuromorphic hardware, which is designed to handle event-driven, spike-based computations

followed by non-linear activation functions (such as ReLU [Nair & Hinton, 2010](#), sigmoid [Turian et al., 2009](#), and tanh [LeCun et al., 2015](#)). To emulate biological neurons, spiking neurons incorporate dynamic properties like membrane potential, spike emission, and refractory periods ([Ghosh-Dastidar & Adeli, 2009b](#)).

Neural activation and information flow: The activation of a neuron model in an ANN is represented by continuous values, and the information is synchronously propagated layer by layer, in other words, through a layer-wise spatial domain ([He et al., 2016](#); [Vaswani et al., 2017](#)). In contrast, the activation of a neuron model in an ANN is represented by discrete spikes. The information is represented and transmitted in an event-driven manner across both temporal and spatial domains ([Wei et al., 2023](#); [Wu et al., 2018b](#)). The event-driven propagation updates postsynaptic neurons only upon spike occurrences ([Wei et al., 2024](#); [Zhang et al., 2021a](#)).

Biological plausibility: While biologically inspired, ANNs are primarily designed to facilitate learning computation ([Taherkhani et al., 2020](#)). In contrast, SNNs offer a more accurate replication of biological neural processes by incorporating mechanisms like membrane potential dynamics and spike generation ([Gerstner & Kistler, 2002](#)). This alignment with biological neurons improves their ability to model complex neural activities with biological interpretation ([Cai et al., 2023a](#); [Yan et al., 2022](#); [Zhang et al., 2022b](#)).

Power consumption: Traditional ANNs convey information through continuous signals, which requires numerous floating-point multiply-add operations, leading to high energy consumption ([Sorbaro et al., 2020](#)). In contrast, SNNs transmit information through binary spikes (0 and 1), involving mostly floating-point addition operations ([Han et al., 2016](#)). Additionally, SNNs operate in a sparse and event-driven manner, where the information is transmitted only upon spike emission, significantly reducing energy consumption ([Sorbaro et al., 2020](#)).

Hardware compatibility: Traditional ANNs were designed for existing computer architecture, such as GPUs and TPUs, optimized for continuous signal processing and parallel computations ([Tye et al., 2023](#)). However, the computational efficiency of SNNs can be realized only on neuromorphic hardware, such as IBM's TrueNorth ([Merolla et al., 2014](#)) and Intel's Loihi ([Davies, Srinivasa, Lin, Chinya, Cao, Choday, Dimou, Joshi, Imam, Jain, Liao, Lin, Lines, Liu, Mathaikutty, McCoy, Paul, Tse, Venkataramanan, Weng, Wild, Yang, & Wang, 2018](#)), which are specialized for handling spike-based, event-driven computations. This neuromorphic hardware offers advantages in energy efficiency and processing speed for sparse and temporal data ([Donati & Valle, 2024](#)).

3. EEG signal and analysis

EEG is a non-invasive method for capturing the brain's electrical activity, providing valuable temporal and spatial information. This section covers the fundamentals of EEG signals, including their physical characteristics, acquisition methods, preprocessing techniques, and feature

extraction. We will also discuss the limitations of traditional ANNs in processing EEG data and introduce SNNs as a promising alternative.

3.1. Brain signals and EEG applications

The brain's ability to deal with complex problems has inspired researchers to study its processing functions and learning mechanisms. This pursuit encompasses various experimental methodologies, broadly categorized into two primary approaches. Hemodynamic measurements, such as functional magnetic resonance imaging (fMRI), positron emission tomography (PET), and functional near-infrared spectroscopy (fNIRS), offer insights into neural activity by detecting changes in blood flow across the entire brain ([Martini et al., 2020](#)). Another is the studies of the activity in brain neurons using various biological signals, including both invasive methods such as Electrocorticography (ECoG) and stereoelectroencephalography (sEEG) ([Mercier et al., 2022](#)), as well as non-invasive methods like electroencephalography (EEG) and magnetoencephalography (MEG) ([Cai et al., 2021](#); [Zhang et al., 2021c](#)). Among these, EEG is the most popular in brain-computer interface (BCI) applications due to its non-invasive nature, convenience, and low cost ([Wu et al., 2022](#)).

EEG signals have provided researchers with profound insights into the brain, driven by advancing technology and research ([Abiri et al., 2019](#)). The applications of EEG-based BCIs are extensive and impactful, enabling motor imagery ([Li et al., 2022](#); [Liao et al., 2023](#)), emotion recognition ([Tan et al., 2021](#); [Xu et al., 2024](#)), cognitive assessments ([Capecci et al., 2015](#); [Doborjeh et al., 2015](#)), and auditory attention detection ([Cai et al., 2021](#); [Faghihi et al., 2022](#)). These diverse applications contribute to our understanding of neural processes, inspiring advancements not only in neuroscience but also in the development of artificial intelligence models that seek to replicate the brain's computational capabilities.

3.2. Fundamentals of EEG signal processing

The analysis of EEG signals involves a series of systematic steps designed to extract meaningful information from raw data. These steps include signal acquisition, preprocessing, feature extraction, and task-specific analysis, as illustrated in [Fig. 3](#).

3.2.1. Physical characteristics of EEG signals

EEG signals are complex, non-linear, and non-stationary, capturing intricate details of brain activity via scalp electrodes. It is essential to understand their physical characteristics so as to design appropriate and effective processing and analysis techniques. The key physical characteristics of EEG signals include:

Waveforms: The amplitude of EEG signals typically ranges from microvolts (μ V) to millivolts (mV) and exhibits diverse waveform patterns, including rhythmic oscillations, sharp waves, and spikes. Sharp waves and spikes often indicate epileptic activity, while rhythmic oscillations

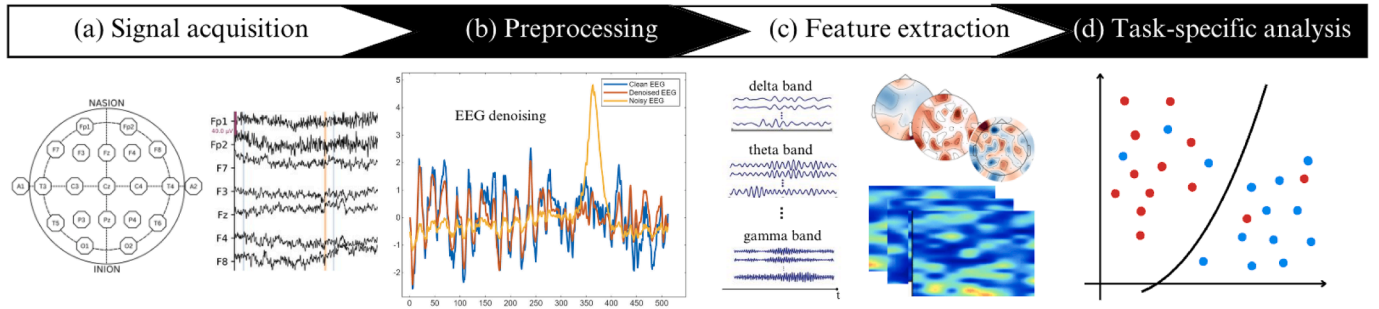


Fig. 3. Overview of EEG signal processing pipeline: (a) EEG signal acquisition (international 10–20 system), (b) Signal preprocessing, (c) Feature extraction, and (d) Task-specific analysis.

are generally associated with states of relaxation or cognitive engagement (Teplan, 2002).

EEG signals typically span a frequency range from 0.5 Hz to 100 Hz (Cai et al., 2022b; Hu et al., 2021a), though higher frequencies (up to 200 Hz) are also recorded in certain contexts, particularly when capturing high-frequency oscillations related to cognitive or pathological activities. The choice of bandwidth depends on the specific application and the nature of the brain activity being investigated.

Standard clinical EEG systems typically sample the EEG signals at 250 Hz to 1,000 Hz, with higher sampling rates used in research context for finer brain responses. Nyquist's theorem suggests that the sampling rate should be at least twice the highest frequency of interest to avoid aliasing (Shannon, 1949). Thus, for standard clinical applications that are interested in bandwidth up to 100 Hz, a sampling rate of 250–500 Hz is commonly sufficient.

Non-linearity: The non-linear properties of EEG signals result from complex interactions among neural oscillations and cortical regions (Stam, 2005). Such interactions give rise to complex patterns that are not easily captured by linear models. Advanced non-linear analyses, such as those using neural networks, have the potential to reveal deeper insights into the underlying brain processes.

Non-stationarity: EEG signals exhibit inherently non-stationary due to their dynamic nature over time (Chen & Ran, 2019; Prochnow et al., 2024). This characteristic presents both opportunities and challenges. Despite the complexity, EEG provides high temporal resolution, enabling the detection of rapid changes in neural dynamics.

3.2.2. EEG signal processing

EEG signal acquisition: EEG signals are acquired through a series of electrodes arranged according to the international 10–20 system as illustrated in Fig. 3(a), which ensures consistent and reproducible electrode placement across subjects (Homan et al., 1987). These electrodes measure voltage fluctuations resulting from neuronal activity, which are then amplified and digitized for subsequent analysis. EEG recordings are susceptible to various artifacts (Dimigen, 2020), including electrical interference, eye movements, muscle movements, cardiac pulsations, etc. These artifacts can obscure the neural signals of interest, complicating data analysis and interpretation. Effective artifact removal is essential for reliable data analysis (Flexer et al., 2005).

Preprocessing: The preprocessing is designed to enhance the digital signal of EEG. Typically, preprocessing techniques include:

- **Filtering:** Bandpass filtering is commonly used in EEG to isolate specific frequency bands of interest and reduce noise. EEG frequencies span several bands, including δ -band (0.5–4 Hz), θ -band (4–8 Hz), α -band (8–13 Hz), β -band (13–30 Hz), and γ -band (30–100 Hz), each associated with different physiological states (Cai et al., 2022b; Hu et al., 2021a). Moreover, notch filters are commonly applied to remove power line interference at 50/60 Hz (Widmann et al., 2015).
- **Artifact removal:** Techniques such as Independent Component Analysis (ICA) are commonly used to identify and remove artifacts

arising from sources such as eye blinks, muscle contractions, and electrical noise (Delorme & Makeig, 2004; Jung et al., 2000).

- **Segmentation:** Continuous EEG data is often segmented into short time windows, each of which is short enough to represent a stable brain state (Kabbara et al., 2023). Such segmentation also facilitates feature extraction for neural network training and evaluation.

EEG characterization: Feature extraction is the process of abstracting relevant information from EEG signals that is relevant to the tasks. It is also called EEG characterization. Next are some common features:

- **Time-domain features:** Neural activities are often reflected as a sequence of temporal events. Basic statistical metrics, such as mean and variance, often fail to capture such temporal information. Recent developments in representation learning offer new ways of abstracting the underlying neural dynamics directly from time-domain EEG signals (Cai et al., 2023a; Sakhavi et al., 2018).
- **Frequency-domain features:** Spectral analysis methods, such as Wavelet Transform (WT) (Faust et al., 2015), decompose EEG signals into their constituent frequencies. Recent studies have further refined frequency-domain analysis by employing neural networks to identify complex patterns and dependencies within spectral data (Wang et al., 2023a).
- **Time-frequency features:** Methods such as Short-Time Fourier Transform (STFT) (Samiee et al., 2015) and Continuous Wavelet Transform (CWT) (Hamaneh et al., 2014) are employed to analyze the temporal and spectral characteristics of EEG signals simultaneously. There are also neural approaches (Madhavan et al., 2020) to capturing how frequency components evolve over time.

Task-specific analysis: While most of the EEG analysis tasks share similar acquisition, preprocessing, and feature extraction frontends, each has a task-specific decision backend. Next are examples of common tasks.

- **Motor imagery:** Motor imagery is a cognitive process that simulates the movement without actual muscle activity. Effective EEG-based motor imagery is crucial in brain-computer interface for rehabilitation and control applications (Liao et al., 2023).
- **Emotion recognition:** In emotion recognition, we analyze how EEG signals are associated with human emotional states, e.g., happiness, sadness, fear, stress (Xu et al., 2024).
- **Diagnosis and cognitive analysis:** By analyzing EEG signals derived from brain activities, we are able to reveal pathological or cognitive patterns, such as diagnosing depression, performing memory tests, or engaging in problem-solving tasks (Doborjeh et al., 2019b).
- **Auditory attention detection:** In this task, we are interested in identifying a listener's auditory attention manifested in his/her EEG signals. It was found that the provoked EEG signals are highly correlated with the attended auditory stimuli (Cai et al., 2024).

3.3. Challenges with spiking neural networks

With the advent of ANNs, we have made progress in EEG analysis by leaps and bounds. Despite notable successes, significant challenges remain in real-world applications.

Traditional ANNs often require large and high-quality datasets for effective training. However, acquiring a substantial amount of such EEG data is particularly more challenging than other data, such as speech or video. EEG data collection is not only labor-intensive, but also relies on sophisticated physical setups. Furthermore, it requires participants to remain focused on specific tasks, which can limit the scale of data collection that can be achieved. As a result, EEG datasets often have a relatively small sample size, typically ranging from 20 to 60 subjects (Chen et al., 2023).

Moreover, traditional ANNs often rely on static, dense feature representations or sequencing of such representations, while EEG signals are instantaneous and sparse with rich temporal dynamics and inherent causal relationships.

For task-specific analysis, traditional ANNs often require extensive computational resources to manage the high dimensionality of EEG data. The complexity of EEG signals, compounded by the need for high-performance models, can strain computational resources, affecting model scalability and real-time application.

SNNs offer a promising alternative. Due to their event-driven nature, they are less dependent on large datasets, making them particularly advantageous in resource-limited EEG scenarios (Cai et al., 2024; Tavanaei et al., 2019). SNNs use sparse, event-driven coding, which closely emulates the way of communication between biological neurons. Therefore, they are particularly suitable for processing instantaneous and nonstationary EEG signals. In addition, the temporal precision and potential for energy-efficient computation also make them a compelling choice for EEG analysis.

4. Spiking neural processing of EEG signals

We now move on to study the key components of spiking neural processing of EEG signals to further understand how SNNs unleash their potential. Preprocessing techniques, such as bandpass filtering and artifact removal, are typically applied before EEG characterization or neural encoding to ensure signal quality, as well as overcome noises (Cai et al., 2023a).

4.1. EEG characterization

The research question is how SNNs abstract relevant information from EEG signals in a similar way to traditional ANNs do. The process of EEG characterization is also called feature extraction. The process typically involves the transformation of data from a high-dimensional to a low-dimensional space, where the low-dimensional representation retains key latent attributes of the original data, ideally closely related to its intrinsic dimensionality (Chen et al., 2024a). SNNs offer distinctive advantages over ANNs for EEG feature extraction due to their biological plausibility and temporal processing capabilities.

First, SNNs use spike trains to transmit information between spiking neurons, in a way similar to that between biological neurons; therefore, more suitable for interpreting brain activities in EEG signals (Gong et al., 2023). Second, SNNs are good at modeling the temporal dynamics inherent in time series, e.g., EEG signals (Cai et al., 2023a). Furthermore, SNNs typically exhibit sparse activation. Only a small number of neurons are active at any given time. The sparsity results in low computational complexity and improved efficiency for feature extraction (Cai et al., 2024). Typically, spiking approaches to feature extraction include:

- **Handcrafted features:** There have been effective handcrafted features in signal processing with known physical properties. One way is to utilize them to characterize EEG signals. For example, in an

auditory attention detection task (Cai et al., 2022a), it was found that spectral analysis serves as an effective feature extraction frontend to the network. The handcrafted features are typically multi-dimensional continuous values that are to be encoded into spike trains by a neural encoder.

- **Representation learning:** This is also called a time-domain end-to-end approach where a learnable network is used to learn the representation directly from raw EEG signals. The representation learning module takes raw EEG signals as input and generates spike trains for subsequent processing. It also serves as a neural encoder. For example, Gong et al. (2023) used multi-head adaptive graph convolutional networks (GCNs) to capture spatial relationships between EEG channels and incorporated spike-based long short-term memory (LSTM) units to capture temporal dependencies, improving emotion recognition accuracy.

4.2. Encoding of spike trains

An encoder serves as the frontend to convert either EEG signals or continuous values, such as handcrafted features, into spike trains. While EEG signals resemble spike trains, they are time series of digitized continuous values. Characterizing the input signals' amplitude and frequency, the spiking neural encoder converts the EEG signals into discrete spike patterns that the subsequent spiking neurons expect (Cai et al., 2024; Gong et al., 2023; Xu et al., 2024).

One common encoding approach is rate coding, where the frequency of spikes corresponds to the magnitude of the input signal (Choi, 2024). Alternatively, temporal coding captures precise timing information of spikes to encode signal dynamics (Zhou et al., 2021). The selection of the encoding scheme depends on the specific application and the desired balance between computational efficiency and biological fidelity (Auge et al., 2021). For real-time implementation, the encoder is expected to handle noisy and high-dimensional EEG signals with low latency.

4.3. Task-specific analysis

Brain-inspired SNNs have emerged as a promising solution to EEG analysis. In Table 2, several representative studies are summarized to provide an overview of the state-of-the-art.

4.3.1. Motor imagery classification

Motor imagery (MI) is an active topic in BCIs, where individuals imagine movements without physically performing them (Li et al., 2022; Nuntalid et al., 2011). Studies showed that motor imagery shares neural pathways with actual motor actions (Lotze & Halsband, 2006), which can be used to develop BCI devices to assist individuals with disabilities, promote rehabilitation for patients with brain disorders, and help prevent motor fatigue (Altafheri et al., 2023; Frolov et al., 2012). SNNs have been increasingly used in MI classification.

A spiking neural network trained with the particle swarm optimization technique was studied for EEG channel selection, which improved classification accuracy by focusing on the most relevant neural signals (Carino-Escobar et al., 2016). Niranjani and Sivachitra (2017) utilized SNNs for EEG-based motor imagery classification through an innovative online meta-neuron based learning algorithm. This method integrated both local spike patterns and global network information to update synaptic weights, showing superior performance (overall testing accuracy: 72.4%) to other classifiers on the planning and relaxed dataset (Blake & Merz, 1998). Mashford et al. (2017) further advanced the practical application of SNNs by designing a two-channel EEG signal classifier compatible with IBM's neuromorphic TrueNorth chip. Using a modified back-propagation algorithm with ternary weights, their approach achieved 84% accuracy on the BCI competition III dataset (Blankertz et al., 2006). This study emphasized the significance of hardware-aware design so as to exploit the full potential for low-latency, energy-efficient real-time motor imagery classification. Li et al.

Table 2

Overview of spiking neural networks from various EEG analysis tasks.

Tasks	Neuron	Feature extraction	Architecture ^c	Datasets ^a	Performance
Motor Imagery	LIF ^b	No	Spike-LSTM (Li et al., 2022)	a	65.75 %
	LIF	Yes	SGLNet (Spiking GCN & LSTM) (Gong et al., 2023)	a	84.15 %
	IZ	Yes	OVR-FBCSP-SNN (Wang et al., 2020)	e	90.09 %
	IZ	Yes	PSD/DWT SNN (Carlos D. Virgilio et al., 2019)	c	81.33 %
				c	88.89 % [†]
	IF, LIF	No	DepthwiseConv (Liao et al., 2023)	b	90.96 %
				c	88.20 %
	LIF	Yes	NeuCube (Behrenbeck et al., 2018)	d	85.18 %
				d	75.00 %
				f	72.40 %
Emotion Recognition	LIF	No	2-channel SNN on TrueNorth (Mashford et al., 2017)	g	84.00 %
	IZ	Yes	PSO-SNN ^d (Carino-Escobar et al., 2016)	h	70.80 %
	LIF	Yes	EESCN (Spiking CNN) (Xu et al., 2024)	i	94.81 %
				j	79.65 %
	LIF	Yes	SGLNet (Spiking GCN & LSTM) (Gong et al., 2023)	i	65.05 % in arousal
				i	59.67 % in valence
	LIF	Yes	Fractal-SNN (Li et al., 2023)	k	69.61 % (arousal), 69.84 % (valence)
				k	78.50 % (arousal), 71.01 % (valence)
	LIF	No	NeuCube BI-SNN (Alzhrani et al., 2021)	j	68.33 %
				l	42.23 %
Disease Diagnosis & Cognitive Task	LIF	No	NeuCube BI-SNN (Alzhrani et al., 2021)	k	94.83 %
	LIF	No	NeuroSense (eSNN & NeuCube) (Tan et al., 2021)	i	78.97 % (arousal), 67.76 % (valence) ^{†f}
	LIF	No	NeuroSense (eSNN & NeuCube) (Tan et al., 2021)	m	79.30 % (arousal), 72.12 % (valence) ^{†f}
				m	75.00 % on Delta
	LIF	Yes	NeuCube (Doborjeh et al., 2019b)	n	81.00 % on Theta
				n	91.00 % on Alpha
	LIF	Yes	Personalised NeuCube (Doborjeh et al., 2019a)	n	85.00 % on Beta
				n	80 % to 93 %
	LIF	Yes	NeuCube (Doborjeh et al., 2018)	n	N.A. ^e
	LIF	Yes	Personalised NeuCube (Doborjeh & Kasabov, 2016)	n	93.61 %
Auditory Attention Detection	LIF	Yes	NeuCube (Doborjeh et al., 2015)	n	86.04 %
	LIF	Yes	NeuCube (Capecci et al., 2015)	n	Eyes closed: 86.00 %
	LIF	No	NeuCube (Kasabov & Capecci, 2015)	o	Eyes open: 79.00 %
	LIF	No	SGCN (Spiking GCN) (Cai et al., 2024)	p	68.70 %
				q	81.40 %
	LIF	Yes	BSAnet (Spiking temporal attention) (Cai et al., 2023a)	p	79.80 %
				q	90.30 %
	IF	No	Probabilistic Spiking Neurons (Faghihi et al., 2022)	q	90.00 %
				q	90.00 %

^a Datasets: a. PhysioNet (Goldberger et al., 2000); b. BCI II-III (Blankertz et al., 2004); c. BCI IV-2a (Tangemann et al., 2012); d. BCI IV-2b (Tangemann et al., 2012); e. BCI III-3a (Blankertz et al., 2006) f. UCI Planning and Relaxed Dataset (Blake & Merz, 1998); g. BCI III (Blankertz et al., 2006); h. BCI III-IVa (Blankertz et al., 2006); i. DEAP (Koelstra et al., 2012); j. SEED-IV; k. DREAMER (Katsigiannis & Ramzan, 2018); l. MPED (Song et al., 2019); m. MAHNOB-HCI (Soleymani et al., 2012); n. Self-collect datasets; o. Dataset for Five Mental Tasks; p. DTU dataset (Fuglsang et al., 2018); q. KUL dataset (Das et al., 2020). Note that the Self-collect datasets differ from each other.

^b LIF = Leaky Integrate and Fire, IF = non-leaky Integrate-and-Fire, IZ = Izhikevich.

^c In Architecture, the main layers used in the neural network are listed. GCN = Graph convolutional network, LSTM = Long short-term memory, Conv/CNN = Convolutional neural network, OVR = One-vs-rest strategy, FBCSP = Filter bank common spatial pattern, PSD = Power Spectral Density, DWT = Discrete Wavelet Transform, OMLA = Online meta-neuron learning algorithm, eSNN = evolving spiking neural network.

^d PSO: Particle Swarm Optimization

^e N.A. : Not available

^f † indicates the performance in subject-independent mode.

(2022) developed a four-layer SNN to address a four-class motor imagery classification problem. In this work, Ben's Spiker Algorithm was introduced to convert EEG signals into spike trains, and a spiking LSTM module was introduced to perform classification. While the proposed method achieved a moderate accuracy of 65.75 % on the PhysioNet data set (Goldberger et al., 2000), the study highlighted the potential of SNNs for motor imagery classification.

Feature extraction techniques also play a key role in EEG-based motor imagery classification. Behrenbeck et al. (2018) introduced the NeuCube, a 3D neurogenetic brain cube designed to mimic the human brain to capture spatial-temporal features, for classifying motor imagery movements from EEG and hand poses from sEMG signals. This approach validated the effectiveness of SNNs and their promise in real-time neuromorphic implementation. Using EEG data from the BCI competition IV (Tangemann et al., 2012), Carlos D. Virgilio et al. (2019) classified

four motor imagery tasks (left hand, right hand, foot, tongue) plus a rest condition using two feature extraction methods, Power Spectral Density (PSD) and Discrete Wavelet Transform (DWT), respectively. In a comparative study with traditional classifiers, e.g., K-Nearest Neighbors and Multilayer Perceptron, spiking neural networks demonstrated superior performance in most scenarios. Wang et al. (2020) studied filter bank common spatial pattern (FBCSP) for spatio-temporal feature extraction in spiking neural networks for multi-class motor imagery classification. After feature optimization using the F-score, 90.09 % and 81.33 % accuracy were achieved on BCI competition datasets III-3a (Blankertz et al., 2006) and IV-2a (Tangemann et al., 2012), respectively. This study highlighted the role of feature extraction in the SNNs approach. Recently, Gong et al. (2023) developed a learnable neural encoder to convert raw EEG signals into spike trains. They employed a multi-head adaptive graph convolution module to capture the spatial features of

EEG signals, alongside a spiking LSTM module to model the temporal dependencies of spikes. The learnable neural encoder acts as a feature extractor. It was reported a superior accuracy of $84.15\% \pm 0.58$ on the PhysioNet dataset. Liao et al. (2023) proposed an SNN, called SCNet, which incorporated a convolutional neural network for representation learning, and achieved good results. SCNet reduces information loss by adaptive coding with learnability and addresses the training difficulty by surrogate gradient learning. SCNet is biologically motivated to be structurally close to the biological neuronal dynamical system. In motor imagery research using ANNs, MixNet (Autthasan et al., 2024) applies the FBCSP method to decompose EEG signals into multiple frequency bands for spectral-spatial feature extraction, achieving a classification accuracy of 77.35 ± 14.41 on the BCIC IV 2a dataset. Yu et al. (2024) enhanced the traditional EMD framework by proposing the Deep-EMD algorithm, which attains a maximum accuracy of 64.19% on the PhysioNet dataset. These comparisons underscore the superior performance of SNN methods.

These studies all revealed the growing potential of SNNs in the real-time processing of EEG signals. By improving upon traditional spiking neural architectures with novel techniques, such as adaptive coding, surrogate gradient learning, and graph convolution, the studies validated the efficiency and effectiveness of SNNs for motor imagery classification.

4.3.2. Emotion recognition

EEG-based emotion recognition is an enabling technology for new BCI applications. It seeks to model the high-level cognitive functions manifested in EEG signals (Tan et al., 2021). It is known that emotions are expressed in physical signals, such as facial expressions, gestures, speech, as well as physiological signals, such as EEG and electrocardiography (ECG), galvanic skin response (GSR), blood volume pulse (BVP), and respiration rate (RT). As physical signals can be easily disguised (Xu et al., 2024), emotion recognition in physiological signals, such as non-invasive EEG, becomes an obvious choice (Li et al., 2019).

There have been successful attempts to apply SNNs in EEG-based emotion recognition. For instance, Tan et al. (2021) proposed NeuSense with an evolving SNN (eSNN) under the NeuCube architecture. This technique, evaluated on the DEAP (Koelstra et al., 2012) and MAHNOB-HCI (Soleymani et al., 2012) datasets in subject-independent mode, achieved classification accuracy of 78.97% (arousal) and 67.76% (valence) on DEAP, and 79.39% (arousal) and 72.12% (valence) on MAHNOB-HCI. Extending the NeuCube framework, Alzhrani et al. (2021) developed a recurrent SNN, NeuCube BI-SNN, for emotion recognition. This technique, evaluated on the DREAMER (Katsigianis & Ramzan, 2018) dataset, achieved 94.83% accuracy in subject-dependent classification of four emotional types, outperforming traditional machine learning methods. Both studies were implemented in the NeuCube architecture, which approximates the structural and functional mapping of human brain areas involved in emotion processing.

Some SNNs are designed to enhance EEG feature extraction. Li et al. (2023) developed Fractal-SNN, which incorporates multiscale temporal-spectral-spatial features, demonstrating effectiveness across four public benchmark datasets. Gong et al. (2023) further tested the proposed SGLNet, which extracts spatial-temporal features, on DEAP dataset (Koelstra et al., 2012). Similarly, Xu et al. (2024) developed Emo-EEGSpikeConvNet (EESCN), which combines neuromorphic data generation with the NeuroSpiking framework. By converting EEG signals into a 2D frame format, EESCN facilitates spatio-temporal feature analysis, achieving subject-dependent classification accuracies of 94.81% on the DEAP dataset (Koelstra et al., 2012) and 79.65% on the SEED-IV dataset (Zheng et al., 2018). The recent progress points to a direction that better performance benefits from more effective feature extraction.

4.3.3. Disease diagnosis and cognitive task

Effective diagnosis and evaluation of the state of clinical subjects is a key aspect of healthcare services. EEG provides a direct measure for

diagnosing brain diseases and analyzing cognitive tasks, exploiting the brain's electrical signals to identify distinctive patterns associated with neurological or psychiatric disorders and to evaluate cognitive functions (Guan et al., 2022; Jiang et al., 2021; Popa et al., 2020).

SNNs offer several advantages, including compact representation in space and time, as well as efficient data learning. For example, the NeuCube architecture (Kasabov, 2014) was specifically designed to model EEG data and uncover functional brain mechanisms associated with cognitive activities. Specifically, NeuCube is an SNN architecture designed for spike timing dependent plasticity (STBD), which consists of several key modules: an input encoding module, a 3D SNN cube, an output classification module, and optionally, a gene regulatory network and optimization module. Kasabov and Capecchi (2015) conducted a case study using NeuCube to model EEG data for cognitive tasks, applying it to a five-mental-task dataset (Keirn & Aunon, 1990). Additionally, Capecchi et al. (2015) employed NeuCube to analyze brain function across three distinct groups: individuals with opiate addiction, patients undergoing methadone maintenance treatment (MMT), and healthy controls. This approach allowed for the comparison of functional brain activity across these groups and the tracking of treatment-related changes, thereby enhancing our understanding of both EEG data and the underlying brain processes. Furthermore, Doborjeh et al. (2015) applied the NeuCube architecture to EEG data collected during a GO/NOGO cognitive task performed by the same three groups: individuals with opiate addiction, MMT patients, and healthy controls. In this study, NeuCube achieved an average accuracy of 86.04% in distinguishing between Go and NOGO tasks. Doborjeh et al. (2018) later integrated dynamic clustering of spatiotemporal data within the NeuCube framework to analyze the same cognitive tasks across the three groups. They also introduced personalized NeuCube frameworks that incorporated individual clinical static data (Doborjeh et al., 2019a; Doborjeh & Kasabov, 2016). In a more recent study, Doborjeh et al. (2019b) applied NeuCube to explore brain function in depression and assess the effects of Mindfulness Training (MT) on both depressed and non-depressed individuals.

Overall, these studies suggested that SNNs, especially with the NeuCube architecture, are effective at modeling the spatio-temporal dynamics of brain activity, offering improved interpretability of EEG signals, therefore, better insights into cognitive tasks and neurological disorders. It was acknowledged that SNNs offer a valuable tool in both research and clinical applications.

4.3.4. Auditory attention detection

Individuals with normal hearing can selectively focus on the speaker of interest in multi-speaker environments, a capability referred to as the *cocktail party effect* (Cherry, 1953). However, hearing-impaired individuals often struggle to identify the speech of interest in such noisy conditions (Cai et al., 2024). Recent advances in neuroscience have shown that neural patterns underlying auditory attention can be identified through the analysis of brain activity, such as ECoG (Mesgarani & Chang, 2012), MEG (Akram et al., 2016), and EEG (Cai et al., 2021; O'sullivan et al., 2015). This is termed *auditory attention detection* (AAD), which opens new possibilities for the development of neural interfaces, including neurally oriented hearing aids (Ceolini et al., 2020).

SNNs have emerged as a promising computational framework for AAD research. Their inherent properties, such as low latency and energy efficiency, align well with the demands of real-world applications in AAD technology. For instance, Faghihi et al. (2022) introduced an SNN model that effectively decodes neural patterns corresponding to leftward and rightward auditory attention, enabling effective training for AAD tasks. Similarly, Cai et al. (2023a) proposed BSANet, which employs biologically plausible spiking neurons and an attentional mechanism to capture the temporal dynamics of EEG signals for AAD. Both models were extensively evaluated on publicly available datasets, demonstrating competitive performance in low-latency settings. These advancements underscore the efficacy of SNNs in modeling the temporal patterns inherent in EEG signals, highlighting their potential in AAD

research and applications. Furthermore, Cai et al. (2024) investigated the use of Spiking GCN to extract spatial features from multichannel EEG, demonstrating promising results in low-density EEG scenarios. Collectively, these efforts highlight the growing role of SNNs in advancing AAD methods and their practical implementation. In representative ANN-based studies on auditory attention decoding, Jiang et al. (2022) introduced the MBSSFCC method, which modifies the traditional SSF-CNN by integrating CNN and ConvLSTM to extract spatiotemporal features from sequences constructed based on multiband topographic activity maps. On the KUL dataset, MBSSFCC achieved accuracies of 86.5 ± 7.16 and 89.5 ± 6.74 under 1-s and 2-s time windows, respectively. On the DTU dataset, the accuracies were 75.6 ± 6.55 and 78.7 ± 6.75 for the same time windows. In contrast, the SNN-based BSA-net (Cai et al., 2023a) method demonstrated superior performance with accuracies of 93.7 ± 4.02 and 95.2 ± 3.08 on the KUL dataset, and 83.1 ± 6.75 and 85.6 ± 6.47 on the DTU dataset for the corresponding time windows, reaffirming the advantages of SNNs in EEG signal analysis.

4.4. Interpretability and explainability

Interpretability refers to the ability to understand or make sense of how a model works, while explainability refers to the ability to provide clear and understandable explanations for its predictions or decisions. As computational models are adopted in practical applications, interpretability and explainability become increasingly important, particularly in medical diagnostics and BCIs (Teng et al., 2022).

ANNs typically operate with continuous activation functions in a dense architecture. It can be challenging to trace the individual contributions to the final output. However, the discrete spiking activity and temporal structure of SNNs provide a more granular view of data processing, facilitating a deeper understanding of the model's behavior. SNNs potentially offer interpretable and explainable solutions (Doborjeh et al., 2021a). By emulating the dynamic processes in human brain more closely than other neural architectures, we hope to understand the underlying neural processes through SNNs (Neftci et al., 2019). This is achieved through discrete spike trains, akin to neuronal firing patterns, which makes the models inherently more interpretable (Maass, 1997). According to Taherkhani et al. (2020), this biological plausibility enhances the intuitive understanding of how SNNs encode and process information, thereby improving their interpretability for EEG analysis. Recent studies further suggest that incorporating brain-like properties (e.g., cross-modal distillation Zhang et al., 2022a or delay modeling Sun et al., 2025a) enhances both interpretability and neuroscientific relevance.

The event-driven nature also makes SNNs more explainable. SNNs operate using sparse, event-driven representations where only significant spikes trigger neuronal responses (Pfeiffer & Pfeil, 2018). In EEG analysis, with sparse coding, only the most relevant features of the signal are processed. This not only reduces data complexity but also makes it easier to trace which aspects of the signal contribute to the model's decisions (Brette, 2015). The focus on the most salient features aligns well with the human approach to reading EEG signals, where experts typically focus on identifying specific patterns or anomalies in the signals (Chaddad et al., 2023).

Additionally, the interpretability of SNNs is enhanced through advanced visualization techniques that provide insights into the network's behavior. Tools such as raster plots (Goloso et al., 2021), which display spike times across neurons, and synaptic weight matrices, which illustrate the connectivity and strength of synapses, are effective for visualizing SNN activity. These visualizations help demystify the internal workings of the network, allowing users to observe how different stimuli or conditions influence the network's responses (Kim & Panda, 2021). In the context of EEG analysis, such visual tools can help pinpoint which brain regions are most active or how different mental states impact neural activity patterns (Luo et al., 2020).

Overall, SNNs are more interpretable and explainable solutions than ANNs in EEG analysis (Doborjeh et al., 2021a; Luo et al., 2020; Neftci et al., 2019; Pfeiffer & Pfeil, 2018). The spike-based processing offers a transparent, event-wise account of information flow, making them uniquely suited for applications requiring both high performance and actionable explanations.

5. EEG-spiking-network project

To put what we discussed above into practice, let's walk through the *EEG-Spiking-Network* project step by step.

5.1. Project overview

The *EEG-Spiking-Network* project, available at <https://github.com/i-spoon/EEG-Spiking-Network>, was developed as a comprehensive framework for classifying EEG signals. It includes scripts for data preparation, spike encoding, spiking neural architecture design, and network training, all implemented in a Python-based environment compatible with deep learning frameworks such as PyTorch. The key features of the project include:

- **Data preparation:** Robust preprocessing scripts are available for organizing raw EEG data into structured training and testing datasets. This process includes data cleaning, normalization, and segmentation.
- **Encoding of spike trains:** Converting continuous EEG signals into discrete spike sequences optimized for SNN processing through a variety of spike encoding paradigms.
- **EEG characterization:** Spectral analysis is performed to extract frequency-domain features, which are critical for identifying underlying neural patterns and improving the performance of the classification model.
- **Task-specific model:** A spiking convolutional neural network (S-CNN), a spiking long short-term memory network (S-LSTM), and a spiking transformer model (S-Transformer, based on the work of Yao et al., 2024a) have been developed for and adapted to EEG signal classification. S-CNN focuses on extracting spatial features from EEG data, while S-LSTM is designed to capture temporal dependencies. S-Transformer employs a spiking attention mechanism and global context modeling to enhance classification performance, particularly for complex and multiscale patterns.

5.2. Data preparation

EEG signals are dynamic time series that manifest the collective activity of neurons. These signals are characterized by two dimensions: the number of channels, representing the electrodes used during data acquisition, and the sequence length, which corresponds to the temporal duration of the recording. Efficient handling of EEG data, including reading and visualization, is crucial for accurate analysis and interpretation. The MNE library (Gramfort et al., 2013) is integrated into this project to facilitate the efficient pre-processing of EEG data, thereby supporting subsequent analysis and task-specific applications.

5.3. Encoding of spike trains

Spike encoding is a process to get continuous signals, either time-domain EEG signals or extracted EEG features, into the SNNs. Effective encoding allows us to capitalize on the low-power advantages of SNNs. Several spike encoding paradigms were implemented in this project, that includes time-to-first-spike (TTFS) (Johansson & Birznieks, 2004), phase coding (Kayser et al., 2009; Pan et al., 2021a), population coding (Averbeck et al., 2006), and the Ternary Asynchronous Encoding (TAE) method (Wang et al., 2024a). These integrated paradigms offer users flexibility in selecting the most suitable encoding technique for their specific applications.

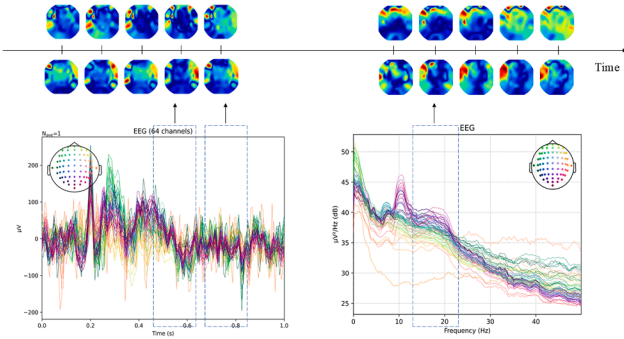


Fig. 4. Spectral analysis of EEG signals.

5.4. EEG characterization

Spectral analysis is widely adopted for EEG characterization. Neuroscience studies have shown that distinct brain activities are linked to specific EEG frequency bands, such as alpha, beta, and gamma rhythms (Anjos et al., 2024; Huang et al., 2024). By incorporating spectral analysis techniques, this project enables users to perform spectral decomposition and analyze these frequency components to gain insights into the underlying brain activity. An example of EEG signal visualization in the frequency domain is shown in Fig. 4.

Spectral decomposition also serve as the features for EEG classification. In Fig. 5, we illustrate a typical convolutional SNN architecture, which produces various classification outcomes. Users can easily stack convolutional network blocks to handle tasks of varying complexity. It was reported that the network achieves low-power consumption in classification results. Moreover, surrogate gradient methods were implemented for network training that address the non-differentiability issue of spiking neurons' gradients.

5.5. Task-specific analysis

The *EEG-Spiking-Network* project features an end-to-end EEG signal classification framework that can be customized for different task-specific analysis.

Taking the EEG-driven auditory attention detection task as an example, the network processes EEG signals as input and performs binary classification to determine whether the attended speaker is on the left or right, as shown in Fig. 5. The key components that are required in the processing pipeline include EEG signal processing, spike encoding, feature extraction, and classification. The modular design of the system ensures that it can be easily adapted to different EEG datasets and experimental setups.

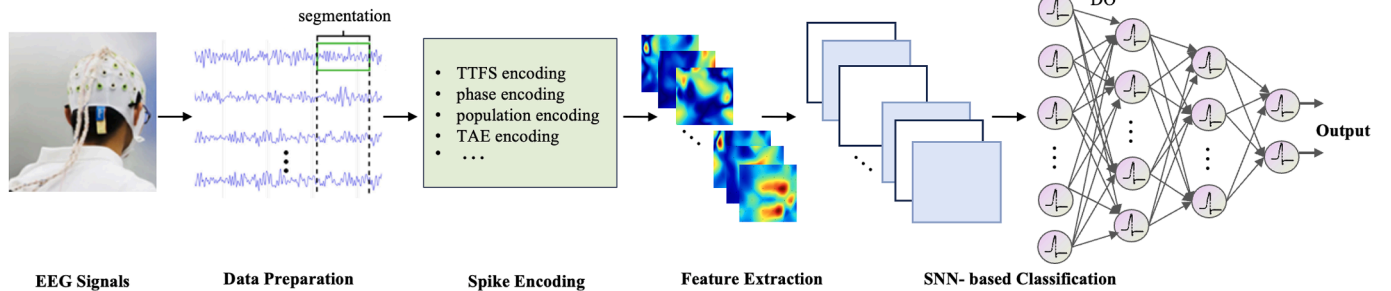


Fig. 5. A spiking neural network pipeline for EEG classification, where EEG signals are first converted into spectro-spatial feature representations, which are then encoded into spike sequences using convolutional spiking layers. The fully connected layer generates the final spike pattern, which serves as the classification output.

The *EEG-Spiking-Network* project come with detailed documentation covering data preparation, spike encoding, spiking neural architecture, training, and evaluation. They are useful resources for people to embark on related research.

6. Future directions and applications

While SNNs offer some advantages in EEG processing, the applications in EEG analysis remain in an early stage. There are both challenges and opportunities. The key limitations, such as information loss from binary spikes, training instability due to surrogate gradients, and inefficient temporal credit assignment, highlight the need for further research. On the other hand, effective neuromorphic deployment of spiking neural networks and joint algorithm optimization on hardware platforms are needed for real-time and low-power applications. These open up ample opportunities for exploration.

6.1. Analyzing the performance gap between SNNs and ANNs

Despite SNNs' efficiency advantages and superior temporal processing capabilities, their task accuracy often lags behind traditional ANNs. This gap stems from three aspects.

First, SNNs transmit information through binary spikes (0 or 1), generated only when membrane potentials exceed a threshold (Gerstner & Kistler, 2002; Golisch & Meister, 2008; Johansson & Birznieks, 2004). This sparse mechanism inherently distorts information during forward propagation, unlike ANNs, which use continuous values, leading to accuracy trade-offs.

Second, SNNs use spikes as the basic units for information transmission, which are discontinuous and non-differentiable. While this enables training, the approximation introduces gradient errors, compromising stability and performance (Li et al., 2021; Meng et al., 2022).

Third, SNNs face a more challenging temporal credit assignment problem than ANNs during error backpropagation. Unlike ANNs, which propagate errors only spatially, SNNs must distribute errors across both spatial and temporal dimensions. This added complexity in tracking errors over time creates significant training difficulties and contributes to SNNs' performance gap relative to ANNs.

6.2. Incorporation of domain knowledge

While SNNs are inherently biologically plausible, their performance and interpretability can be further improved by integrating domain-specific knowledge and neurophysiological constraints.

One key area of improvement involves incorporating realistic neuron models that closely mimic known biological spiking behaviors, enabling SNNs to emulate how neural populations process and transmit information in the brain (Saranirad et al., 2024). By embedding anatomical and

functional constraints—such as brain connectivity patterns and cortical organization—into network architectures, SNNs can better capture the hierarchical and distributed nature of brain activity (Cai et al., 2024). Moreover, explicitly encoding the functional roles of frequency bands (e.g., delta waves in sleep or gamma rhythms in attention) allows SNNs to amplify clinically relevant features while suppressing noise, enhancing the detection of subtle neurological biomarkers (Tawhid et al., 2024; Yu et al., 2023). Such approaches not only boost accuracy but also make the model's decision-making process more transparent to clinicians.

Incorporating biological constraints, such as synaptic plasticity rules and neuronal dynamics, is another critical aspect. STDP and other biologically inspired learning rules have been shown to enhance the learning capabilities and generalization network performance. These rules mimic the natural learning processes observed in the brain, making the models more adaptive and interpretable. For instance, Dong et al. (2023) demonstrated how STDP could be used to train SNNs effectively, improving their ability to learn from temporal patterns in EEG data and offering insights into the underlying neural mechanisms.

6.3. Novel architectures and algorithms

Although SNNs are believed to be suitable for EEG signal analysis, current studies are limited in two aspects: overly simple spiking convolutional architectures and inadequate consideration of power consumption requirements.

The rich spatiotemporal patterns in EEG signals demand sophisticated processing mechanisms. The integration of spatiotemporal attention (Su et al., 2022) could significantly enhance feature extraction, where temporal attention captures dynamic patterns while spatial attention models channel relationships. This weighted filtering strategy is particularly suitable for SNNs, as their event-driven sparsity and adaptive firing thresholds allow for the straightforward implementation of attention through selective spike propagation. Recent work on parameter-free attentional SNNs (Sun et al., 2025b) demonstrates this potential for noise-resilient processing without explicit denoising.

Moreover, graph representation of EEG signals was found to be effective (Cai et al., 2024; Gong et al., 2023). EEG signals can be naturally represented as graph-structured data, where an electrode represents a node and the spatial relationships or correlations between electrodes are edges. Such a graph representation can effectively explore the spatial structure of EEG channels and encode the temporal dependence of EEG spikes, thereby enhancing BCI performance.

Mobile EEG applications call for real-time data processing, long-term monitoring, and small-footprint and low-power implementation. Although SNNs are advantageous due to their event-driven and sparse computing properties. The prior studies have not fully exploited such advantages. This presents a future direction.

6.4. Potential applications

The research on SNNs for EEG signal analysis enables new services and applications and transforms both clinical and non-clinical practices.

Clinical diagnostics: SNNs hold considerable promise for enhancing the diagnosis of neurological disorders due to their adeptness at processing temporal information with high precision. For instance, Zarrin et al. (2020) demonstrated that a deep SNN model achieved 97.6% accuracy in detecting epileptic seizures from the Kaggle intracranial EEG dataset, outperforming ANNs with 95.2% accuracy under identical preprocessing conditions. Burelo et al. (2022) reported that SNNs achieved 15% higher classification accuracy (80%) compared to ANNs in detecting epileptic high-frequency oscillations (HFOs) in noisy scalp EEG recordings. This enhancement is attributed to SNNs' event-driven processing, which improves noise robustness by selectively responding to salient temporal features. Beyond epilepsy, SNNs also enable earlier detection of neurodegenerative diseases such as Alzheimer's and Parkinson's by identifying subtle EEG abnormalities that precede clinical symp-

toms (Doborjeh et al., 2021b). This capability allows for risk prediction in patients with mild cognitive impairment, paving the way for timely therapeutic interventions (Capecci et al., 2016).

Wearable BCI devices: SNNs offer distinct advantages for real-time EEG signal processing on mobile and edge computing platforms with constrained computational resources. They are particularly well-suited for deployment in mobile applications due to their inherent event-driven and sparse computing properties. This allows for effective online processing of EEG signals on a variety of devices, including smartphones, tablets, and wearable BCI devices (Lobo et al., 2020).

Long-term monitoring: SNNs are effective for the long-term monitoring of cognitive states such as stress, fatigue, and sleep stages, which manifest as subtle and transient changes in EEG. By focusing on significant fluctuations, SNNs ensure continuous monitoring and provide valuable insights into biological neural processes. Their real-time processing capabilities support tracking cognitive load and delivering feedback in various settings. SNNs facilitate ongoing brain activity monitoring through compact and energy-efficient components, potentially revolutionizing personal health monitoring and remote patient management with continuous insights.

7. Conclusion

In this review, we advocate that SNNs offer a promising alternative to EEG signal analysis through a large body of literature. SNNs were found to outperform traditional ANNs in terms of capturing and processing the dynamic nature of brain signals. This review also highlights the transformative potential of SNNs in real-time and low-resource implementation. By bridging the gap between theory and practice, we provided comprehensive resources that empower researchers to start related research or to enrich their toolbox. Based on the recent research progress, this review also provides a perspective on future directions of related studies.

CRedit authorship contribution statement

Siqi Cai: Writing – review & editing, Methodology, Validation, Conceptualization, Writing – original draft, Formal analysis; **Zheyuan Lin:** Validation, Writing – original draft, Formal analysis, Writing – review & editing, Methodology; **Xiaoli Liu:** Writing – original draft, Formal analysis, Methodology, Validation; **Wenjie Wei:** Validation, Writing – original draft, Formal analysis, Methodology; **Shuai Wang:** Methodology, Writing – original draft, Formal analysis; **Malu Zhang:** Supervision, Formal analysis, Writing – original draft, Investigation, Writing – review & editing, Resources, Conceptualization; **Tanja Schultz:** Writing – review & editing, Conceptualization, Funding acquisition, Supervision; **Haizhou Li:** Writing – original draft, Investigation, Writing – review & editing, Resources, Conceptualization, Supervision, Funding acquisition.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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